

Desk controller

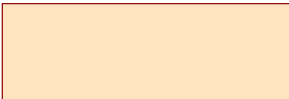
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Change history

1.0.0	07/10/2026	First release
1.0.0	08/10/2026	Fix R1234 value due to issue xyz etc etc etc. Update C14 to 10 nF to reduce roll-off frequency.
1.0.0	TBD	TBD
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System overview



File: System_overview.kicad_sch



Desk controller

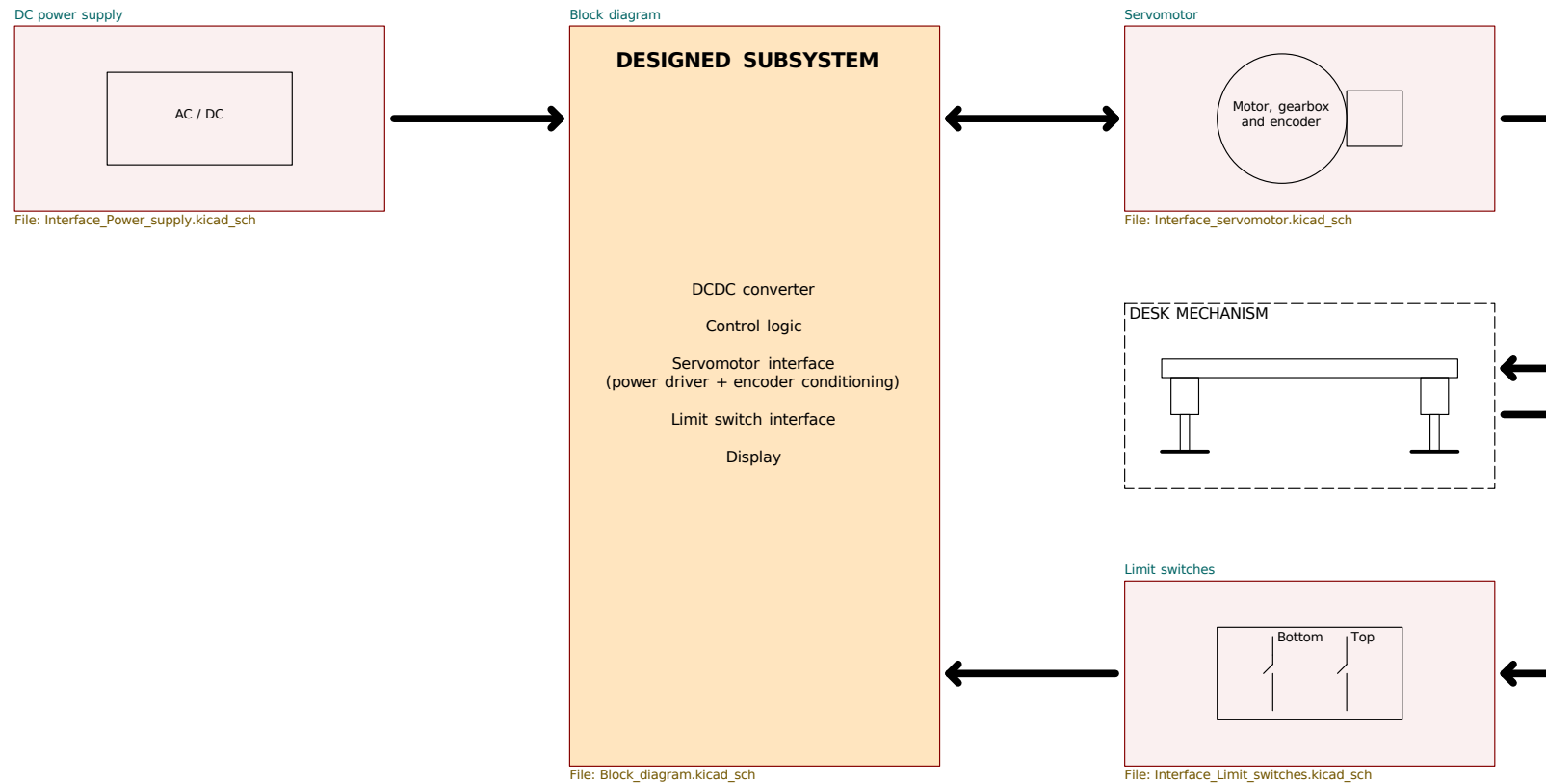
Notes

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This is a comment 2 field - potentially not used
This is a comment 3 field - potentially not used
This is a comment 4 field - potentially not used

Sheet: Root
File: ktest.kicad_sch

Rev: 0.0.1	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date: 2026-10-06	RLs'd: FLR
Dwg: NA	Size: A4
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System overview



This sheet shows the overall system and the interfaces to external components. External components are shown for system context only and are not part of this schematic. Only the "DESIGNED SUBSYSTEM" is covered by this schematic.

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Desk controller

Notes

Sheet: System overview

File: System_overview.kicad_sch

Rev:

Mat: Concept

Date:

Dwg: NA

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Drwn: FLR

Rev'd: FLR

RLs'd: FLR

Size: A4

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DC power supply

Interface specification

Barrel connector (Dell type, with ID pin)

Pin

- 1 Supply - 19.5 V, 3 A (inner shell)
- 2 ID (internal pin)
- 3 GND (outer shell)
- SH Shield

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Desk controller

Notes

Sheet: DC power supply	
File: Interface_Power_supply.kicad_sch	
Rev:	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date:	RLs'd: FLR
Dwg: NA	Size: A4
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Interface specification

JST B6B-XH-A / JST XHP-6 connector

Pin	Description
1	Encoder B output (VCC)
2	Encoder A output (VCC)
3	GND
4	VCC (3.3 - 5V)
5	Motor B (3 A, 12-24 V)
6	Motor A (3 A, 12-24 V)



Desk controller	Sheet: Servomotor	
	File: Interface_servomotor.kicad_sch	
	Rev:	Drwn: FLR
	Mat: Concept	Rev'd: FLR
	Date:	RLs'd: FLR
Notes	Dwg: NA	Size: A4
	Generated with KiCad	Page: 4/11

Limit switches

Interface specification

JST B4B XH A / JST XHP-4 connector

Pin Description

- 1 Bottom limit switch (Normally open, switch closes to GND)
- 2 GND
- 3 GND
- 4 Top limit switch (Normally open, switch closes to GND)

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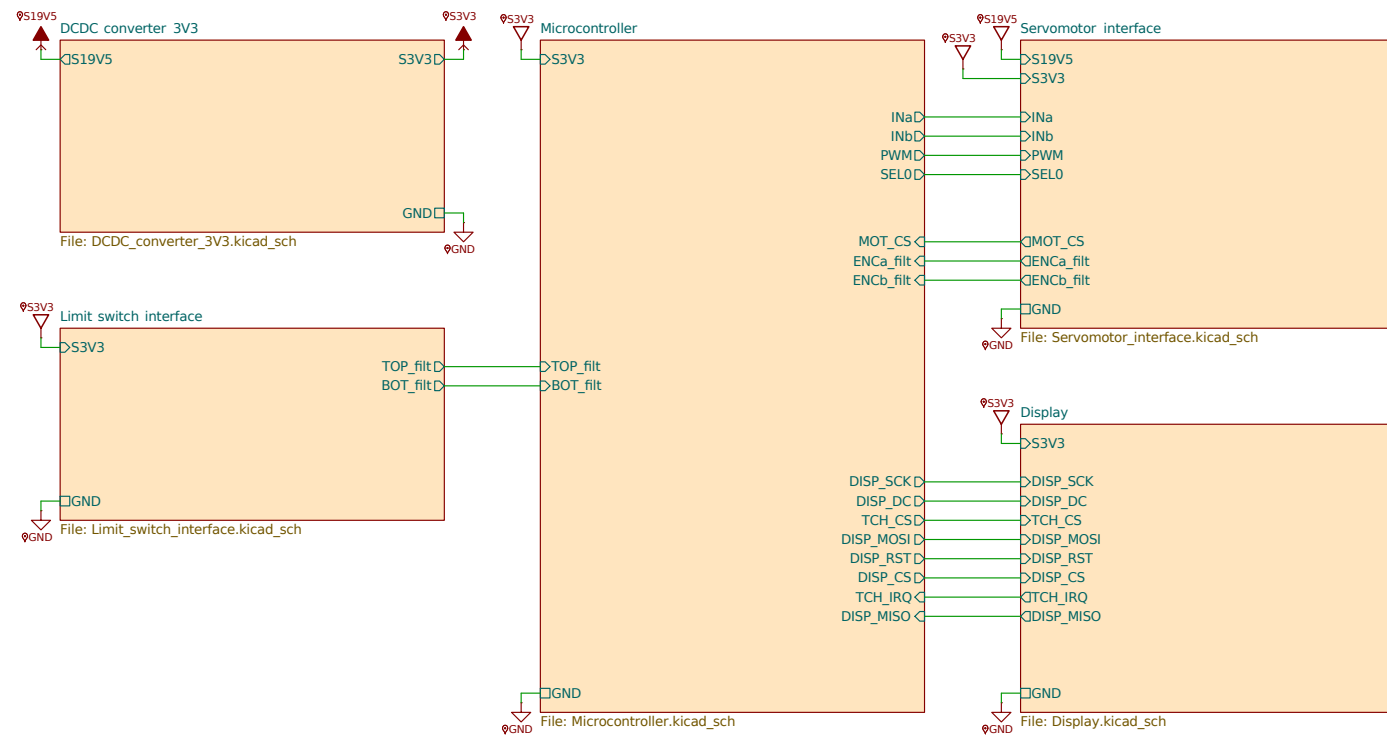


Desk controller

Notes

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Rev:	Drwn: FLR
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Date:	RLs'd: FLR
Dwg: NA	Size: A4
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Block diagram



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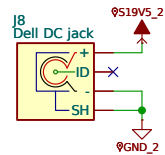
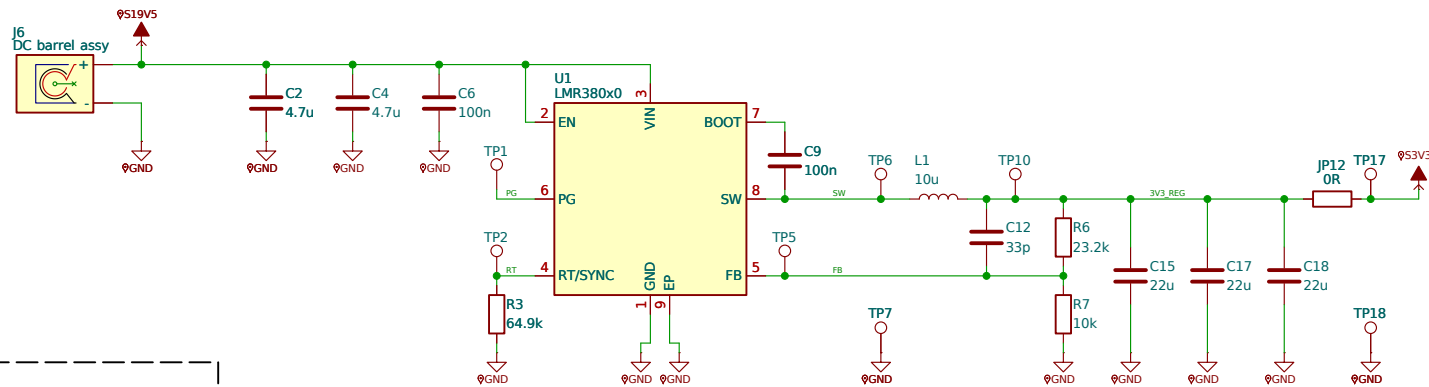
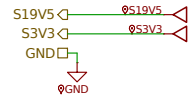
Desk controller

Notes

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Rev:	Drwn: FLR
Mat: Concept	Rev'd: FLR
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Dwg: NA	Size: A4
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DCDC converter 3V3

Block interface



J8 is fitted to a small "standalone" PCB which will then be soldered into the "main" PCB. This standalone PCB + connector are what make the "DC barrel assy" used in J6 position.

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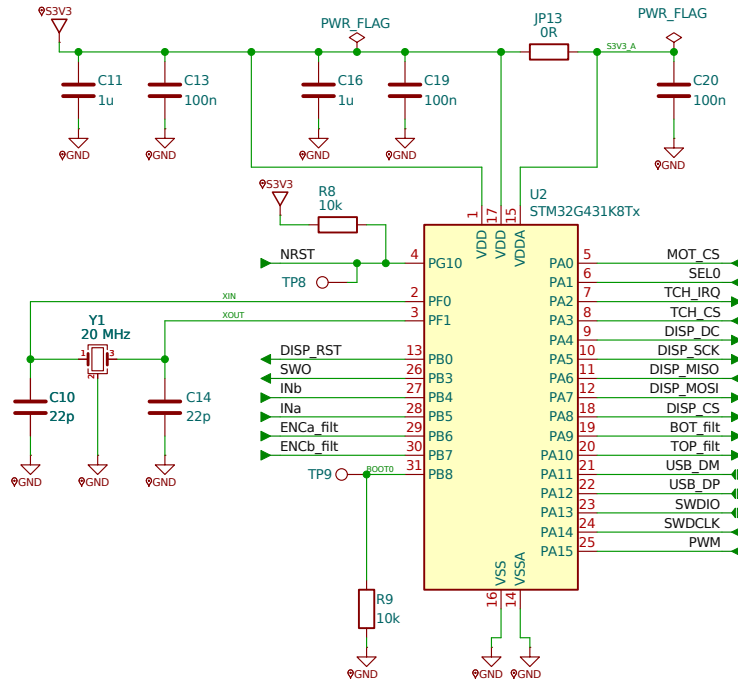
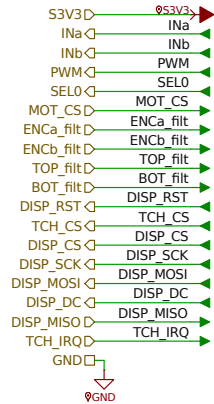


Desk controller

Notes

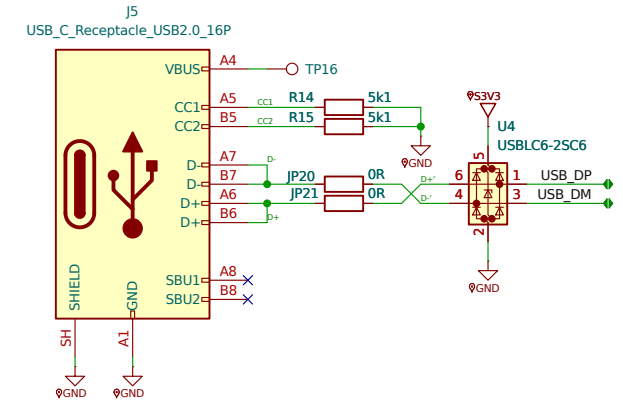
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Date: NA	Rev'd: FLR
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Block interface

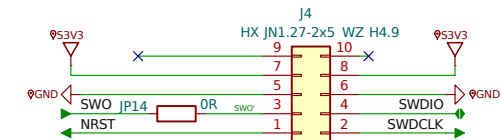


Microcontroller

USB interface (serial comms port)



SWD header



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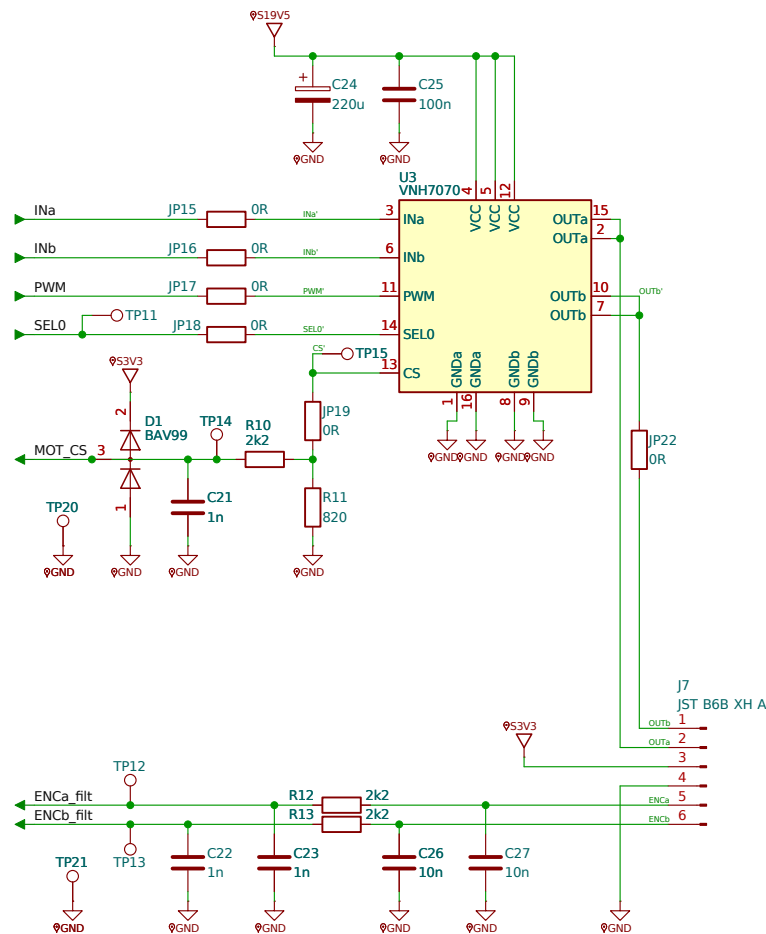
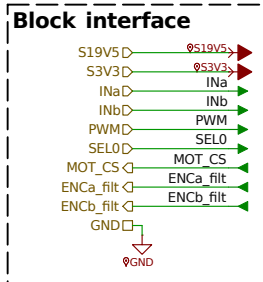


Desk controller

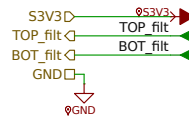
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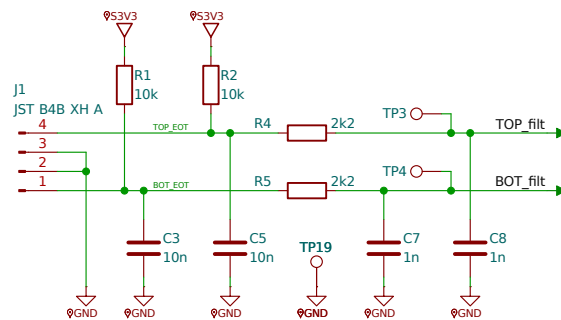
Servomotor interface



Block interface



Limit switch interface



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Desk controller

Notes

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Rev:

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Dwg: NA

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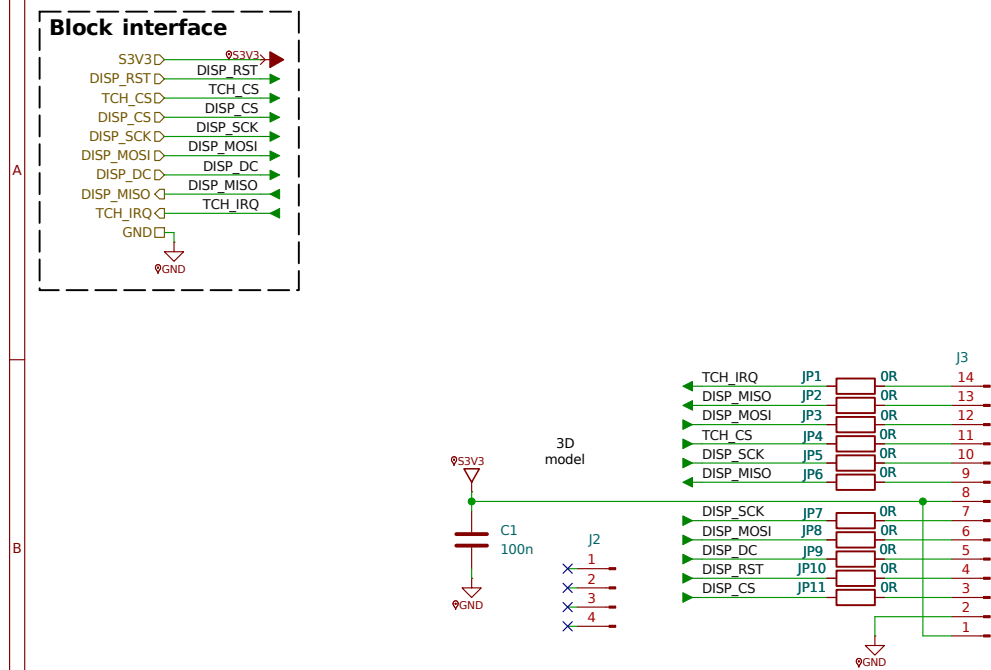
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Display



TFT1
TFT_2.4in_320x240

The display will not be directly soldered to the PCB. Instead, generic 2.54 mm pitch sockets J2 and J3 will be used as connectors.

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